

Understanding Prism-Thinning

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BACKGROUND

In an earlier Tech Talk, we discussed the optics and geometry of *progressive addition lenses (PALs)*. It was pointed out that, in a progressive lens, the curvature of the surface gradually increases toward the bottom of the lens as illustrated in Figure 1. This increase in curvature (and surface power) is what produces the *add power* of a progressive. Unfortunately, because the bottom of the lens is steeper than the top of the lens, the upper edge of a progressive lens blank is thicker than the lower edge.

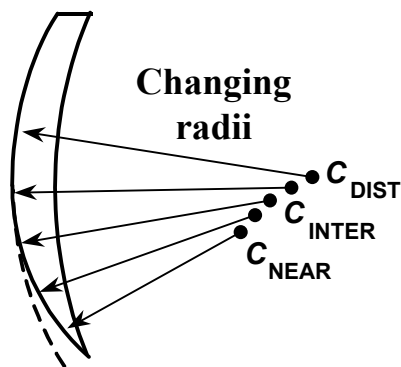


Figure 1. A conventional progressive lens steepens in curvature towards the bottom to produce the desired add power. Notice the difference in thickness between the top and bottom edges.

This lens geometry requires a greater center thickness in order to provide the same minimum edge thickness in the near zone (at the bottom edge). Consequently, for plus-powered lenses—or lenses with a significant add power—a normal progressive lens blank will be thicker than a conventional flat-top lens blank of the same power. This might seem a bit counter-productive for a lens designed to improve cosmetics!

Many dispensers are already aware of the problems inherent with such a lens blank from past experiences with executive-style multifocals. The steeper segment curvature of an executive-style bifocal also creates a thickness differential between the upper and lower edges.

Another consideration is the fitting height of progressive lenses. In the absence of prism, the optical center of a PAL will typically be ground at the **prism reference point (PRP)** of the lens. The PRP is

generally at the geometric center of the lens blank, about 2 to 4 mm below the fitting cross. If a pair of lenses requires an extremely high fitting height, the PRP—and optical center—of each lens needs to be decentered upwards.

For high-powered lenses, this vertical decentration can also result in a noticeable thickness difference between the upper and lower edges of the finished lens. (Interestingly enough, in some cases the thickness difference produced by the fitting height can actually offset the thickness difference produced by the progressive lens geometry!)

Consider Figure 2, which illustrates the edge thickness around the perimeter of a +2.00 DS progressive lens with a +2.00 D add. The lens has a center thickness of 4.19 mm. There is also a 2-mm thickness difference between the upper and lower edges of the finished lens!

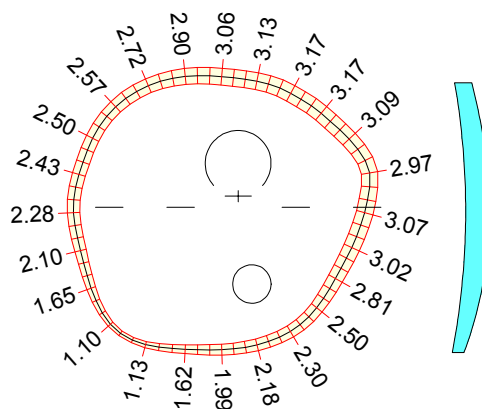


Figure 2. Progressive lens with a +2.00 DS distance Rx and a +2.00 D add. The center thickness is 4.2 mm, and the weight is 9.4 g.

So what can be done to ensure your progressive lens patients receive the superior cosmetics they expect? How can the excess thickness and weight of progressive lenses be minimized? The answer: *prism-thinning*. **Prism-thinning** (also called **equi-thinning**) is the process of grinding prism into a progressive lens blank to reduce the thickness difference between the upper and lower edges.

Prism-thinning typically involves grinding *base down* prism into progressive lenses. In addition to balancing the thickness difference between the top and bottom

of the lens blank, prism-thinning can also reduce the center thickness of progressive lenses with plus power and/or higher add powers.

Prism-thinning is accomplished during the generating process by literally tilting the front surface of the lens on the chuck of conventional generators, using a *prism ring*. When the back surface is ground normally, the surfaced lens is left with a prismatic effect at the center. This process is illustrated in Figure 3.

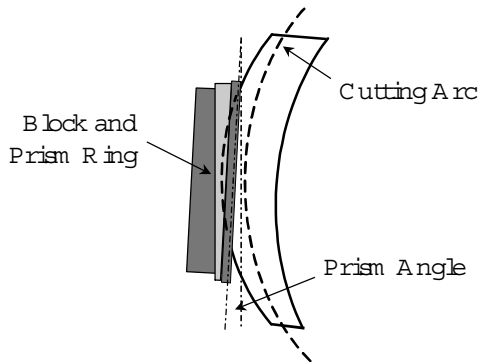


Figure 3. Base down prism is ground into the lens blank using a prism-ring (with its base positioned up) with conventional generators.

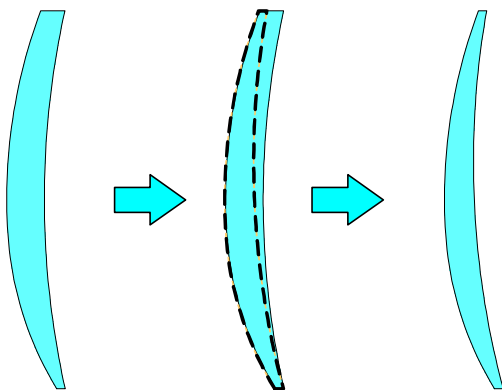


Figure 4. Grinding base down prism into a progressive lens reduces the thickness difference between the top and bottom of the lens blank, as well as some of the center thickness.

Newer, three-axis generators produce this prismatic effect without the use of prism rings by grinding the back curve with a tilt. The end result in either case is a reduction of unwanted thickness, as shown in Figure 4.

To avoid creating a vertical prismatic imbalance, the same quantity of vertical prism is ground in both lenses. This prism is often referred to as *yoked prism*, since no *net* (or binocular) prismatic effect is produced.

So how much prism-thinning should be used? Ideally, the amount of prism-thinning used should be based upon the following factors:

- Distance power (in the vertical meridian)
- Add power
- Fitting cross height
- Fitting cross decentration
- Frame shape

For instance, lenses with higher plus powers in the distance portion require more prism-thinning. This is also true for progressive lenses with higher add powers. The fitting height needs to be considered to take into account the thickness difference produced by vertical decentration, as described earlier.

A common rule-of-thumb formula provided in some progressive lens processing guides is:

$$Prism = 0.6 \times Add$$

This shows that a quantity of base down prism equal to roughly 2/3rd of the add power should be used. This is often recommended when the power through the vertical meridian of the lens exceeds +1.50 D or so. This formula does not consider factors like the fitting height and the distance power, but still produces satisfactory results in most cases.

Consider the prism-thinned lens in Figure 5; this is the same lens used earlier (in Figure 2). With 1.33^A of base down prism-thinning ground in—which is 2/3 the add power—the finished lens now has a center thickness of only 3.53 mm. This is a 16% reduction in thickness from the previous example, which had no prism-thinning. The thickness differential has also been reduced. Moreover, the lens is also significantly lighter in weight (by 19%). Clearly, the prism-thinning has made this progressive lens both thinner and lighter; and by no small amount!

Instead of rule-of-thumb formulas, many laboratories use complex computer programs to determine the correct amount of prism-thinning.

Laboratory software packages from companies like *Digital Vision Inc.* and *Gerber-Coburn*, for instance, give the laboratory several prism-thinning options to choose from. These options include using either a set value or calculating the exact amount of prism thinning-required. When the prism-thinning is computed in this fashion, the exact amount of prism is determined based upon all of the factors described earlier. This method will produce the thinnest possible lens configuration.

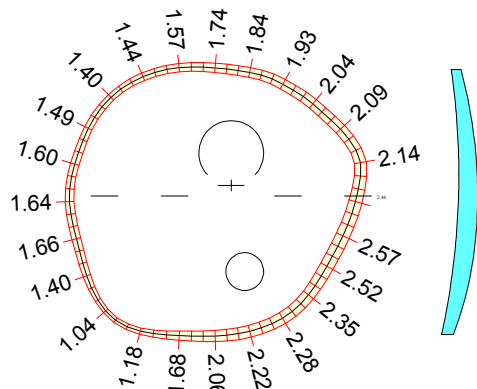


Figure 5. Progressive lens with a +2.00 DS distance Rx and a +2.00 D add. Now, 1.33^Δ of base down prism-thinning has been used—which is 2/3 the add power. The center thickness is 3.5 mm, and the weight is 7.6 g.

Consider the prism-thinned lens in Figure 6; this is also the same lens used earlier (in Figure 2). In this case, the *exact* amount of prism-thinning has been computed. With 1.76^Δ of base down prism-thinning ground in the finished lens the finished lens now has a center thickness of only 3.33 mm.

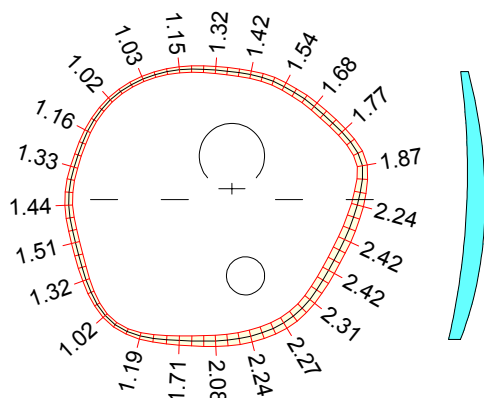


Figure 6. Progressive lens with a +2.00 DS distance Rx and a +2.00 D add. Now, 1.76^Δ of base down prism-thinning has been used. The center thickness is only 3.3 mm, and the weight is 7.1 g.

This represents a 21% reduction in thickness over the example, which had no prism-thinning! The thickness differential has been virtually eliminated. Further, this lens is also 25% lighter in weight!

It is possible to prism-thin minus-powered lenses, as well. Depending upon the fitting height, either *base down* or *base up* prism may be required to balance the thickness difference. Consider the prism-thinned lens in Figure 7; this -5.00 DS progressive lens has a +2.00 D add and a relatively high fitting height. To balance the upper and lower thickness differential, 1.62^Δ of *base up* prism-thinning has been used. This

keeps the lower edge of the lens from looking noticeably thicker than the upper edge.

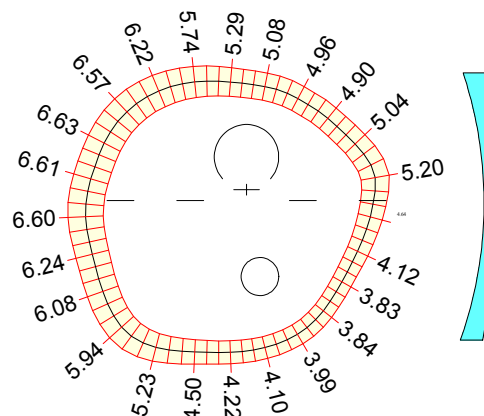


Figure 7. Progressive lens with a -5.00 DS distance Rx and a +2.00 D add. Now, 1.62^Δ of *base up* prism-thinning has been used.

Now that we have discussed why you should use prism-thinning, we need to address how to verify lenses with it. In terms of vertical prism, you are primarily concerned with the *net* vertical prismatic imbalance.

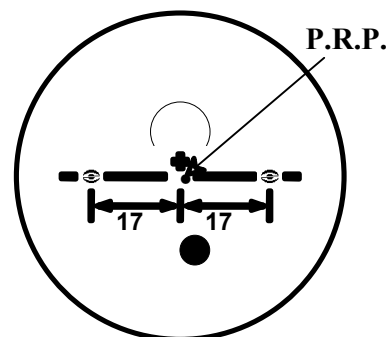


Figure 8. Prism should be verified at the prism reference point (PRP) of a progressive lens, which is centered between the alignment markings (or logos).

Begin by placing the PRP of each lens in the center of the lens stop of your focimeter. The PRP is illustrated in Figure 8.

Verifying the amount of prism-thinning in a lens is especially important when you are replacing only one of the lenses. If the previous lens had not been prism-thinned and the new one was—or vice versa—an unwanted vertical prismatic imbalance will be induced.

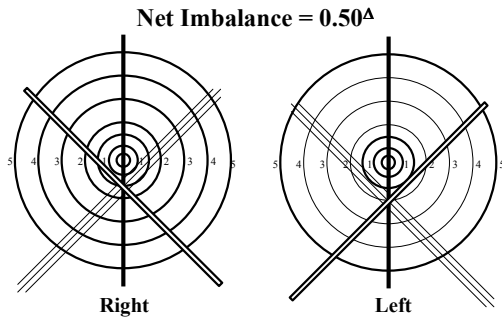


Figure 9. Note the amount of vertical prism at the PRP of each lens. The difference is the amount of vertical prismatic imbalance (or net vertical prism). In this example, the net imbalance is 0.50^{Δ} .

You have seen some pretty big figures used for prism-thinning in these examples. So just how much vertical yoked prism is acceptable? Is there a limit to the amount of prism-thinning that a patient can tolerate? Sheedy and Parsons performed a study a few years ago exploring the effects of vertical yoked prism on patient acceptance. This study showed that a group of test subjects was not significantly affected by 2.00^{Δ} of vertical prism (and no significant postural adjustments were made). However, 4.00^{Δ} of vertical prism was rejected by almost all of the test subjects. Consequently, the limit of prism-thinning for most patients will probably lie between 2^{Δ} to 4^{Δ} .

Winthrop pointed out one interesting consideration for prism-thinning lenses. The *reflected* ghost images created by lenses with prism power are deviated more than the actual *refracted* image. This means that lenses with prism power will produce reflected ghost images that may be noticeably offset from the original object, as illustrated in Figure 10.

What does this mean to a wearer? Certain reflections may be more noticeable. For example, a wearer might suddenly notice the reflection of a brightly lit office window because the reflected image moves up enough to be seen at eye level. Of course, these reflections will be eliminated with anti-reflective coatings—a popular option for any premium lens.

Wearers of high-powered lenses—which naturally induce a prismatic effect as the wearer looks away from the center—are already used to this phenomenon. Wearers of low-powered lenses, on the other hand, may not notice these offset ghost images unless they have prism ground into their lenses. The reflected ghost images may also be slightly clearer (or more in focus) in low-powered lenses. Under these circumstances, prism-thinning might produce ghost images that are distracting enough to the patient to be

an annoyance. Fortunately, this is a very rare occurrence.

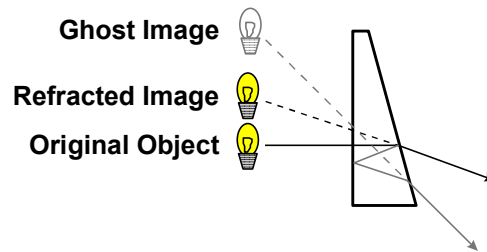


Figure 10. The *reflected* ghost image produced by a prism is deviated significantly more than the *refracted* image of the object.

Certain highly aspherical progressive lenses may not need to be ground with prism-thinning because of the geometry of the surface. Some lenses, like the American Optical *Omni*, are surfaced with ‘shims’ on the front surface, which keep the lens from rocking on the block. Between the aspherical nature of the distance portion and the tilt produced by the shims, the thickness differential is minimized without the need for grinding additional prism. The lenses are automatically produced with prism-thinning.

In summary, prism-thinning is a useful technique that improves both the finished cosmetics and comfort of many progressive lenses—with little (if any) visual impact to the wearer. Although the focus of this article has been on the thickness reduction of progressive lenses, it is equally possible to use prism-thinning for Executive-style multifocals. Recall that they share a geometry similar to progressive addition lenses. Some laboratories may offer prism-thinning as an option for both of these multifocal styles. As a matter of fact, if you have questions about the criteria that your particular laboratory uses to apply prism-thinning (e.g. which lenses, how much, etc.), be sure to give them a call.